2019

YEAR 10 SCIENCE

James Ruse Agricultural High School

YEARLY EXAMINATION

SOLUTIONS

General Instructions

- Reading time – 5 minutes
- Working time – 95 minutes
- Write using black pen.
- Draw diagrams using pencil.
- NESA approved calculators may be used.
- A periodic table of elements provided.
- Write your name at the top of each separate section booklet.

Total Marks – 80

Section I – Biology

20 marks

Section II – Chemistry

20 marks

Section III – Physics

20 marks

Section IV – Earth and Space

20 marks

Section I – Biology

Name: .....

Class: .....

Mark: .....

Answer Grid for Multiple Choice Questions

1.	A O	B O	C O	D O
2.	A O	B O	C O	D O
3.	A O	B O	C O	D O
4.	A O	B O	C O	D O
5.	A O	B O	C O	D O

Part A – Multiple Choice Questions **5 marks**

1. What group of organisms are responsible for the breakdown and recycle of matter?

A Herbivores
B Omnivores
C Detritivores
D Carnivores

2. Which two organ systems regulate and maintain homeostasis?

A The nervous and endocrine systems.
B The cardiovascular and integumentary systems.
C The cardiovascular and respiratory systems.
D The respiratory and muscular systems.

3. Two plants of the same species are crossed by a plant breeder. Both plants are pure breeds for flower colour. All the offspring plants have the same flower colour, but originate from different parents.

What type of inheritance does this example demonstrate?

A Sex-linked.
B Recessive
C Dominance
D Co-dominance

4. Rosalind Franklin was an English Chemist and an X-ray crystallographer.

Her work on DNA contributed to the discovery of which feature?

A Long stranded molecule.
B Double helix shape
C Sugar-phosphate backbone
D Complementary nucleotides

5. Over the past several decades, natural selection has caused populations of *Staphylococcus aureus* (an infectious wound bacterium) to evolve resistance to most antibiotics.

If the usage of antibiotics was stopped, what is most likely to happen to the populations of *S. aureus*?

A They will go extinct without the antibiotic.
B The frequency of resistant forms will increase in these populations.
C The populations will begin colonizing new environments.
D The frequency of non-resistant forms will increase in these populations.

Part B – Extended Response Questions 15 marks

Question 6 (2 mark)

The human body has many lines of defence to combat disease.

Identify the first and second lines of defence. **2**

First: Physical or chemical barriers eg. skin, mucous, hair, pH of stomach.

Second: Innate, non-specific eg. phagocytes, inflammation, blood clotting, fever.

Criteria	*Marks*
• Identifies the line of defence **OR** • Provides several examples of components within the line of defence	2
• 1 of the above	1

Question 7 (3 marks)

A particular species of guinea pigs is known to have either black or white coloured fur, where the black fur (B) is dominant over white fur (b). A homozygous black guinea pig is crossed with a heterozygous black guinea pig.

Construct a punnett square to determine the probability of offspring with the same phenotype as the parents. **3**

	B	B
B	BB	BB
b	Bb	Bb

100 % of offspring will have the same phenotype.

Criteria	*Marks*
ONE mark for each of the following: • Correct Phenotype - 100% black offspring • Correct arrangement of parent genotype in punnett square • Correct arrangement of offspring genotype	3

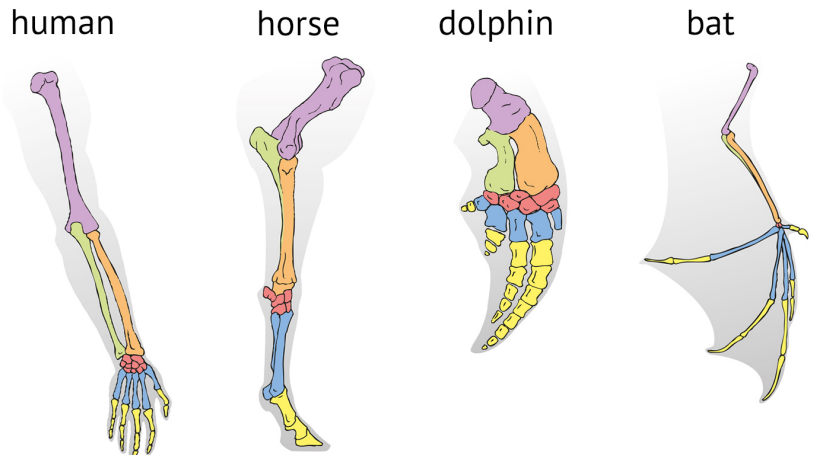
Question 8 (3 marks)

Describe an interspecific interaction between organisms using ONE named example. **3**

Criteria	*Marks*
ONE mark for each of the following: • Interactions between organisms belonging to different species • Accurate named example provided • ONE of the following correctly represented: - Commensalism - Mutualism - Parasitism - Competition - Predation	3

Question 9 (3 marks)

Many living organisms have basic structures with similarities, such as the Pentadactyl limbs shown below:



Describe the term Pentadactyl limb and explain how the similarities in each organism relates to the theory of evolution. **3**

Criteria	*Marks*
• *Penta = 'five', 'dactyl' = finger* Five-fingered limb. • Divergent Evolution (Basic structures show that they have evolved from a common ancestor.) • Homologous structure (The limbs above all look different, with different functions, but each possesses a limb that ends in five fingers.)	3
• TWO of the above	2
• ONE of the above	1

Question 10 (4 marks)

Explain how the following evidence supports the Theory of Evolution.

(a) Fossils **2**

Answers may incluide:

- Fossils provide evidence of progression or changes in organisms over the life of the earth.
- Fossil transitions, fossil indexing & dating
- Fossils provide evidence of extinct species and how they relate to extant species (morphological and anatomical)

Criteria	*Marks*
● Identifies types of fossils OR provides a valid example ● Explains how fossils support the theory of evolution	2
● ONE of the above	1

(b) Geographic variation **2**

Answer may include:

- Distribution of organism across varying geographical locations with different selective pressures (e.g. abiotic factors)
- Ecological niches and changes in genetics flow and genetic drift in conjunction with the movement of tectonic places over geological time.
- Allopatric speciation due to geographic isolation with similar traits or genetic evidence to suggest a common ancestor.

Criteria	*Marks*
● Identifies what geographic variation is **OR** ● Provides a valid example of an organism that has undergone speciation **AND** ● Explains how geographical isolation supports the theory of evolution	2
● ONE of the above	1

End of Section I

Section II – Chemistry

Name: .....

Class: .....

Mark: .....

Answer Grid for Multiple Choice Questions

1.	A O	B O	C O	D O
2.	A O	B O	C O	D O
3.	A O	B O	C O	D O
4.	A O	B O	C O	D O
5.	A O	B O	C O	D O

Part A – Multiple Choice Questions **5 marks**

1. An element on the periodic table is represented using the following notation:

$$^{X}_{Y}Z$$

Which of the following is true?

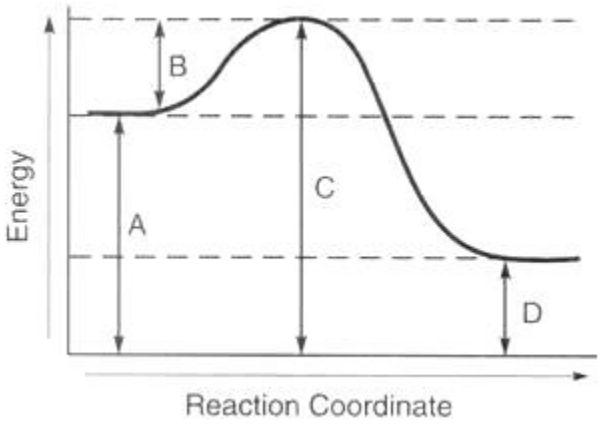
A X is the atomic number
B X is the mass number
C X is the number of protons
D X is the number of neutrons

2. A pure sample of an element is shiny, soft and reacts with oxygen at room temperature.

What group is this element most likely to belong to?

A Alkali metals
B Transition metals
C Halogens
D Noble gases

Use the information provided in the Energy Profile diagram to answer questions 3 and 4.



3. Which of the following statements is true?

A The reaction is exothermic and the reaction mixture would heat up.
B The reaction is endothermic and the reaction mixture would heat up.
C The reaction is exothermic and the reaction mixture would cool down.
D The reaction is endothermic and the reaction mixture would cool down.

4. The rate of reaction is determined by which of the following?

A A
B B
C C
D D

5. Which of the following is an example of a neutralisation reaction?

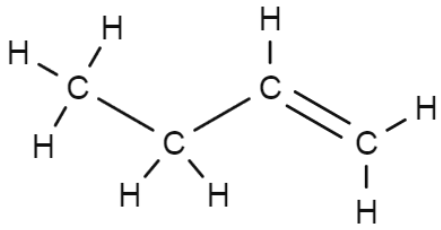
A Hydrogen ions and electrons forming hydrogen atoms.
B Hydrogen chloride reacting with ethylene to produce chloroethane.
C Hydrochloric acid reacting with lead nitrate to form solid lead chloride.
D Hydrochloric acid reacting with magnesium hydroxide to produce magnesium chloride and water.

Part B – Extended Response Questions 15 marks

Question 6 (3 marks)

Nomenclature allows chemists to communicate the structure and names of molecules.

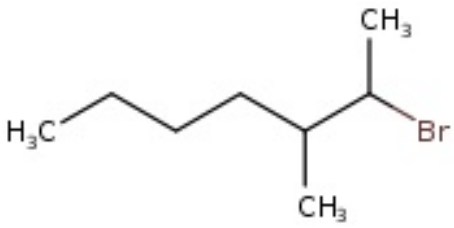
(a) Name the following molecule: **1**



But-1-ene (1-butene accepted)

1 mark	Correctly names structure

(b) Draw the structure for a molecule of 2-bromo-3-methylheptane in the space provided. **2**



2 marks	Draws correct structure
1 mark	Draws structure with one or two errors

Question 7 (3 marks)

Write a balanced chemical equation, with states, for the reaction of nitric acid with calcium carbonate chips to produce calcium nitrate, carbon dioxide, and water. **3**

$$2HNO_{3\,(aq)} + CaCO_{3\,(s)} \rightarrow Ca(NO_3)_{2\,(aq)} + CO_{2\,(g)} + H_2O_{\,(l)}$$

3 marks	Correct chemical formulae Balanced equation Correct states
2 marks	2 of the above OR All of the above with up to 2 errors
1 marks	1 of the above OR All of the above with up to 4 errors

Question 8 (2 marks)

Outline the similarities and differences between the bonding present in MgO and CO. **2**

This question referred to the bonding, not the formula or the physical properties of the compounds.

Similarities: The bonds in both compounds involve using valence electrons to try to form a stable outer shell. They are chemical bonds and require a substantial amount of energy to break.

Differences: In MgO, there is an ionic bond with electrostatic attraction between the negatively charged oxygen and positively charged magnesium. In CO there is a covalent bond with some valence electrons shared between the carbon and oxygen atom.

2 marks	Outlines one similarity and difference between the bonding
1 mark	Outlines one similarity or difference in bonding

Question 9 (2 marks)

For ONE named polymer, describe a property that is related to its use. **2**

High density polypropylene (HDPE) is resistant to many solvents and is used as a container for fluids such as drinks and petrol.

2 marks	Identifies one polymer Describes one property Relates property to common use
1 mark	Identifies one polymer Identifies one property or common use OR Identifies class of polymers Describes on property Relates property to common use

Question 10 (2 marks)

A student places a piece of zinc metal into a solution of copper sulfate and makes observations.

(a) Describe ONE observation. **1**

Answers may include:

The zinc would become coated in a red-brown metal.

The copper sulfate solution would become less blue.

The zinc electrode would decrease in mass.

1 mark	Describes possible observation

(b) Identify whether the observation is the consequence of an oxidation half reaction or reduction half reaction. **1**

Answers may include:

This is due to copper ions from the solution combining with electrons from the zinc to form copper metal, a reduction reaction.

This is due to copper ions leaving the solution and combining with electrons from the zinc to form copper metal, a reduction reaction.

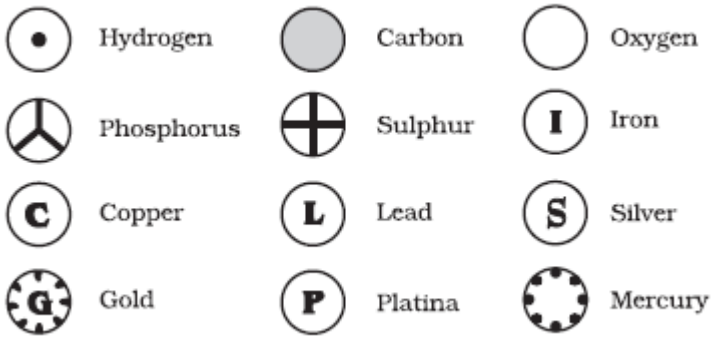
This is due to the zinc atoms giving electrons to the copper ions, and the zinc ions going into solution, an oxidation reaction.

1 marks	Explains observation in part (a) in terms of redox reactions

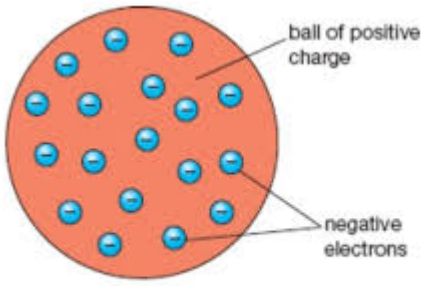
Question 11 (3 marks)

Outline the contribution of ONE named scientist to the development of atomic theory. **3**
Include a labelled diagram of their atomic model in your response.

Around the time of Dalton (early 1800s) , an understanding of different substances and the ratios in which they combined was being studied. He proposed that substances were made of tiny particles called atoms. Atoms were indivisible particles, with each element having its own type, that all atoms of one element were the same and that elements combined in fixed whole-number ratios to form compounds. In chemical reactions, atoms were rearranged. He determined relative atomic masses for six elements and some compounds. His atoms did not have a defined structure, but he gave symbols to them:

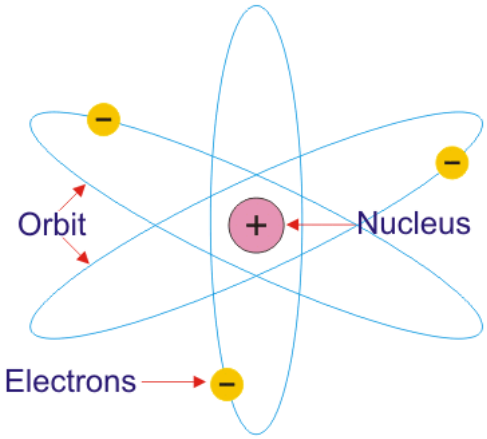


In 1897 Thomson discovered the electron, demonstrating that atoms contained small negative particles and were therefore not indivisible. In 1904 he proposed a model in which a large diffuse positive mass was studded with small negative charges, like raisins in a plum pudding, giving rise to the term "plum pudding model".

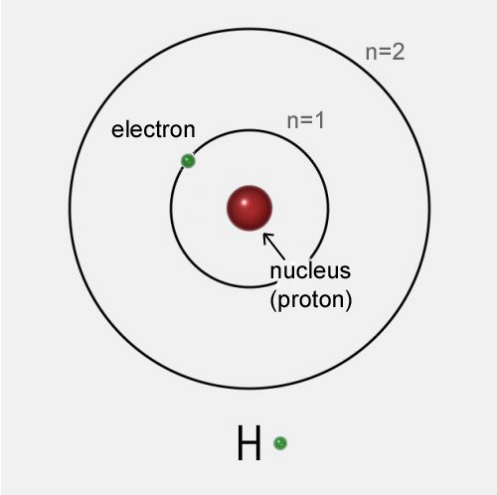


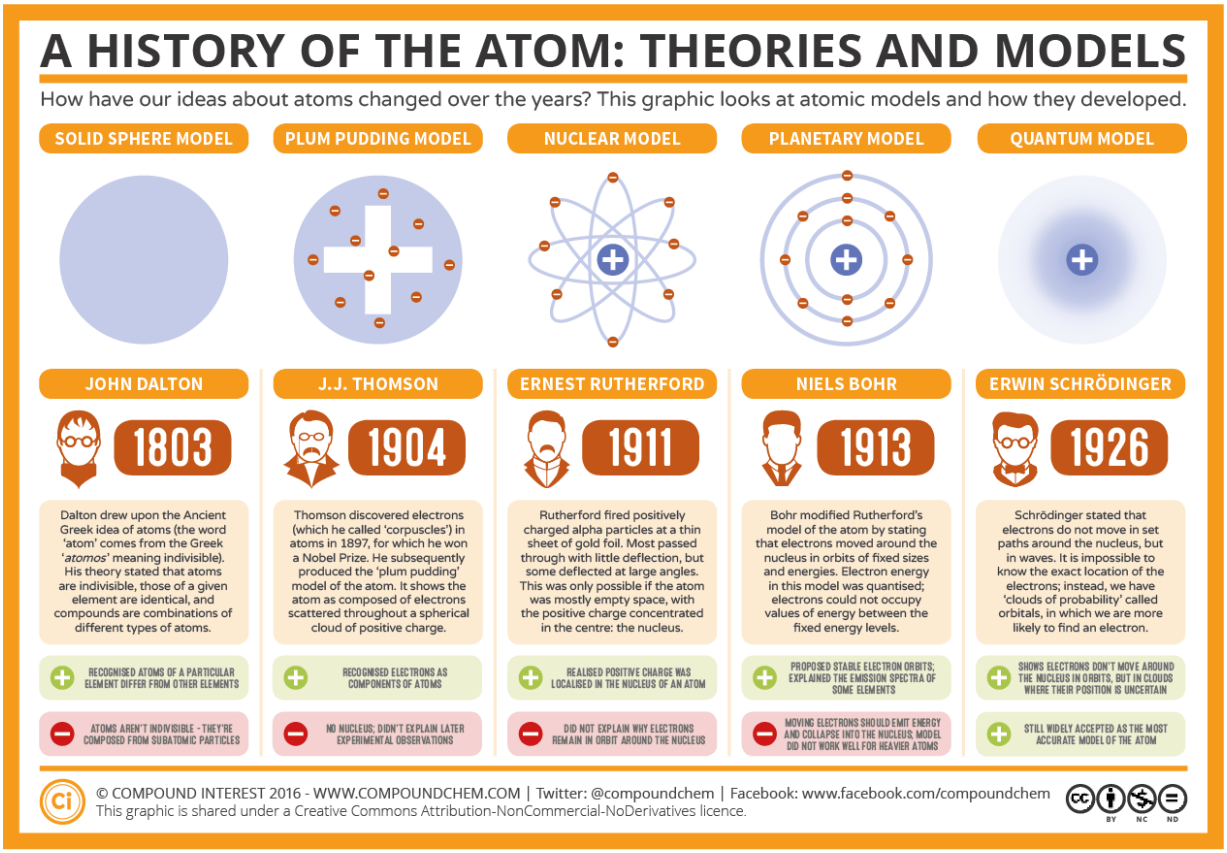
Thomson's 'plum-pudding' model of the atom

In 1909, he used the Geiger-Marsden experiment in which alpha-particles were fired into gold foil to search for ones which were deflected by a large angle. Some were found, suggesting that there was a large, dense particle repelling the positively charged alpha particle. His 1911 model of the atom had a positive, dense mass at the centre (the nucleus) and smaller electrons orbiting the nucleus, in mostly empty space. In 1920 he suggested that the hydrogen nucleus was a proton, and in 1921 he proposed the existence of neutrons.



In 1913, to overcome the physical impossibility of charged electrons orbiting the nucleus without emitting radiation and decaying inwards, Bohr proposed a model in which electrons could only exist in certain orbits called shells without radiating energy. Electrons could only move between these shells by emitting or absorbing EMR, which could account for spectral lines seen in emission spectra.





3 marks	Names relevant scientist Identifies reason(s) for change to atomic theory Identifies change(s) to atomic theory Draws labelled diagram of atomic model
2 marks	3 of the above
1 mark	2 of the above

End of Section II

Section III – Physics

Name: .....

Class: .....

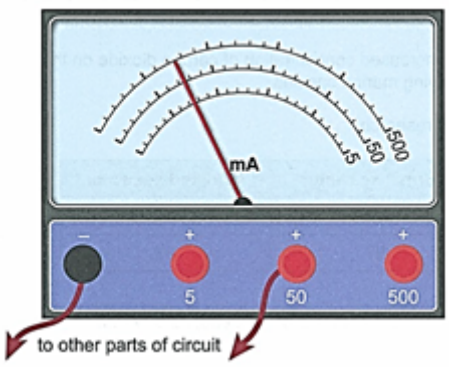
Mark: .....

Answer Grid for Multiple Choice Questions

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3.	A O	B O	C O	D O
4.	A O	B O	C O	D O
5.	A O	B O	C O	D O

Part A – Multiple Choice Questions **5 marks**

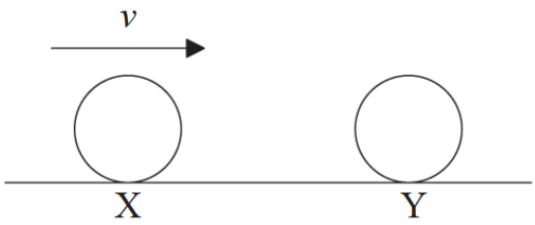
1. The diagram provided shows an electrical ammeter with multiple scales. This ammeter has been connected to a circuit with current flowing.



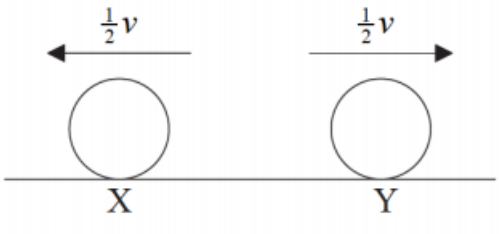
What is the reading (in mA) shown on the ammeter?

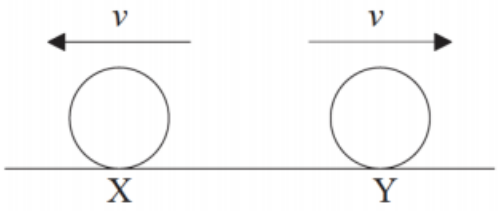
A 1.5
B 12.5
C 15
D 125

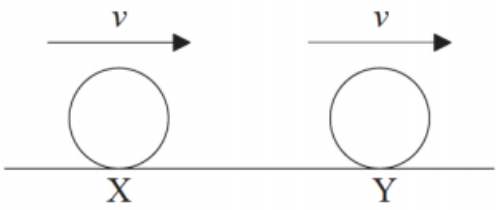
2. A simulation is setup where ball X is moving on a frictionless surface in a horizontal line to collide with an identical ball Y that is at rest.

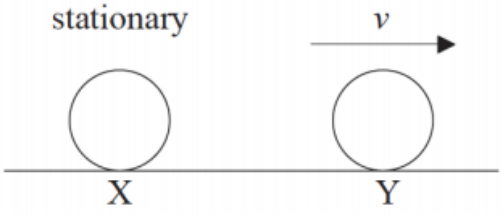


When ball X strikes ball Y with speed v, what is a possible outcome of the collision?

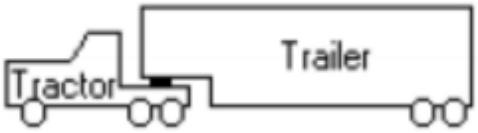
A 

B 

C 

D 

3. A tractor-trailer truck is travelling down a straight road. The tractor has a mass of m, which is connected to a trailer with a mass of 4m.



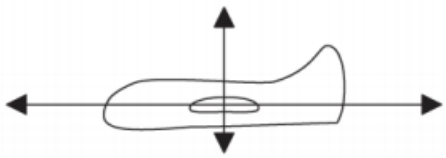
m **4m**

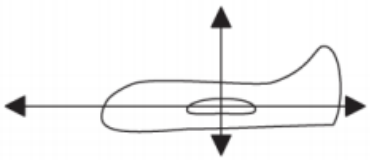
What is the force applied by the trailer to the tractor, when the tractor accelerates forward?

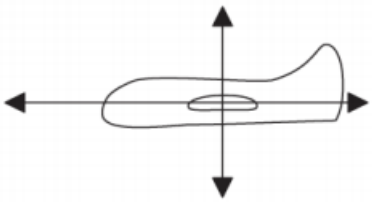
A ¼ the force of the tractor on the trailer.
B equal to the force of the tractor on the trailer.
C 4 times the force of the tractor on the trailer.
D 5 times the force of the tractor on the trailer.

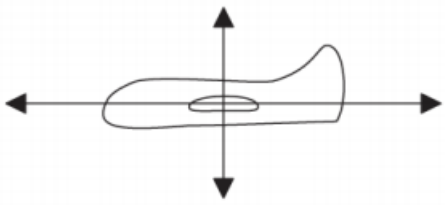
4. A model plane flies with constant velocity at a constant height.

Which diagram represents the forces acting upon the plane?

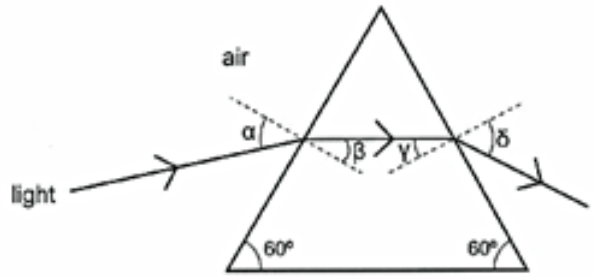
A 

B 

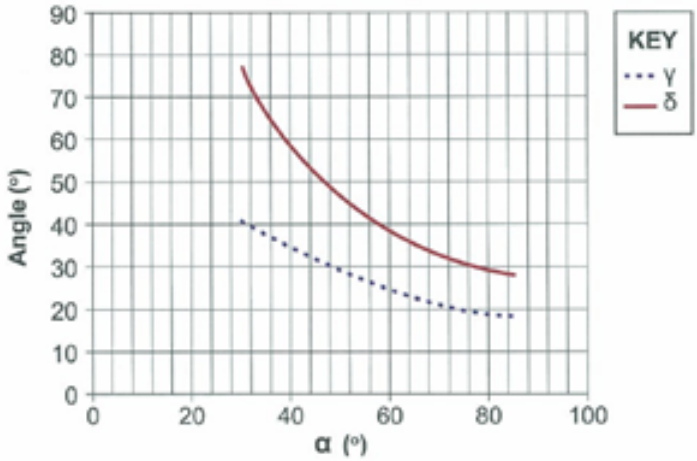
C 

D 

5. Anne drew a diagram to show the path a ray of light would take through a prism.



She varied the angle of incidence (α) whilst measuring the other angles each time and graphed her results.



As she moved the light source in an arc to vary α, she made sure that the light ray struck the prism at the same point on the surface.

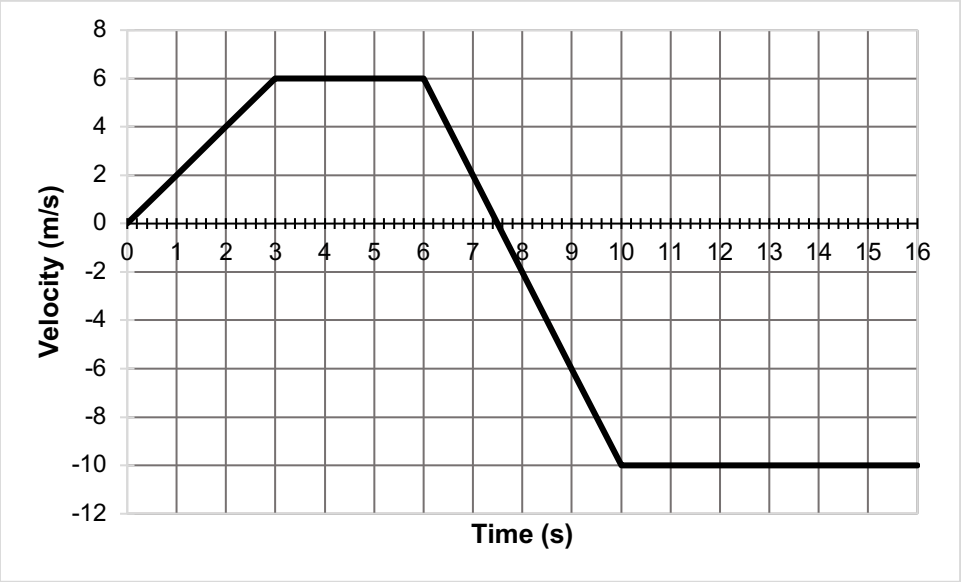
Which of these is an observation that Anne could have made during her investigation?

A The angle α is always larger than the angle γ.
B An increase in angle β produces an increase in angle γ.
C The light rays always leave the prism at the same point.
D The angle δ can reach a value of 90 degrees but the angle γ cannot.

Part B – Extended Response Questions 15 marks

Question 6 (4 marks)

Shown below is a velocity-time graph for a motorcyclist's journey.



(a) Calculate the total distance travelled by the motorcyclist. **2**

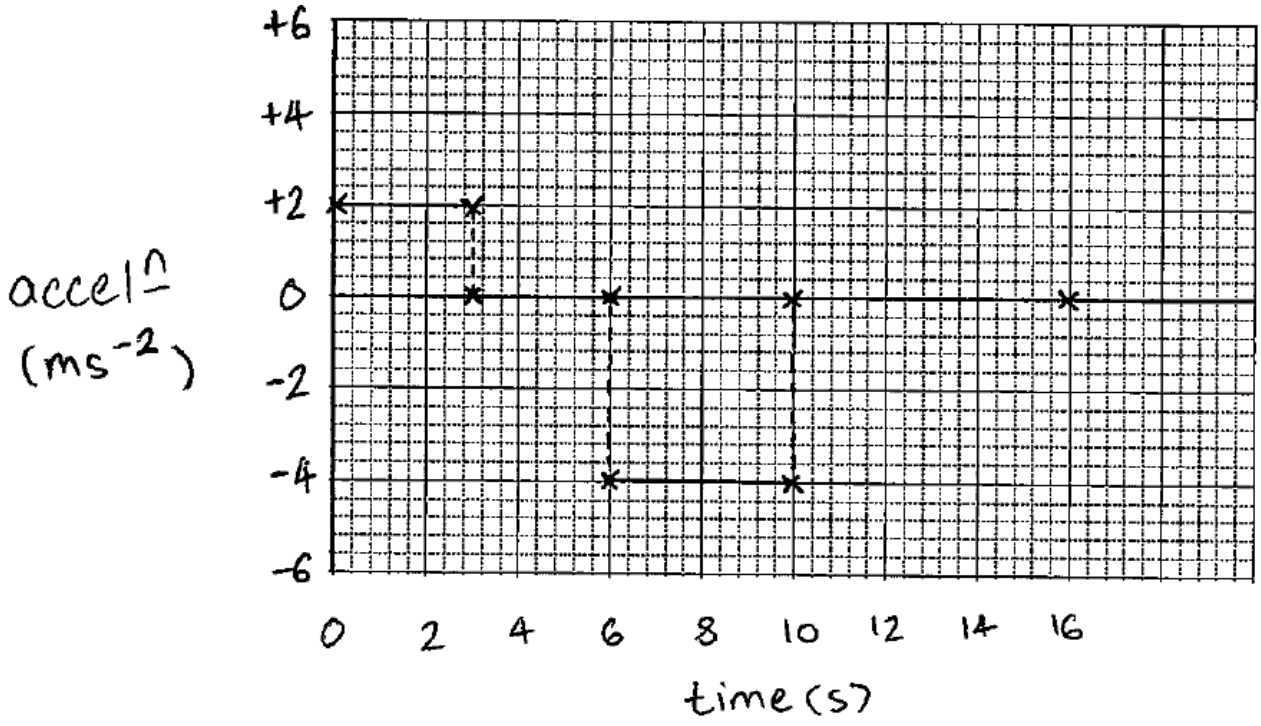
Distance = (6 x 3 / 2) + (6 x 3) + (6 x 1.5 / 2) + (10 x 2.5 / 2) + (10 x 6)

= 9 + 18 + 4.5 + 12.5 + 60

= 104 m

Marking Criteria	*Marks*
• Correctly calculates the total distance travelled. • Provides relevant working out.	2
• Provides some relevant working out (partially correct calculation for area under the graph).	1

(b) Draw an acceleration-time graph of the motorcyclist's journey on the grid provided. **2**



Marking Criteria	*Marks*
• Correctly draws an acceleration – time graph • Provides correct axes labels with an appropriate scale.	2
• Correctly draws an acceleration – time graph **OR** • Correctly plots / draws moments of constant acceleration (0 to 3 seconds and 6 to 10 seconds). • Provides correct axes labels with an appropriate scale.	1

Question 7 (4 marks)

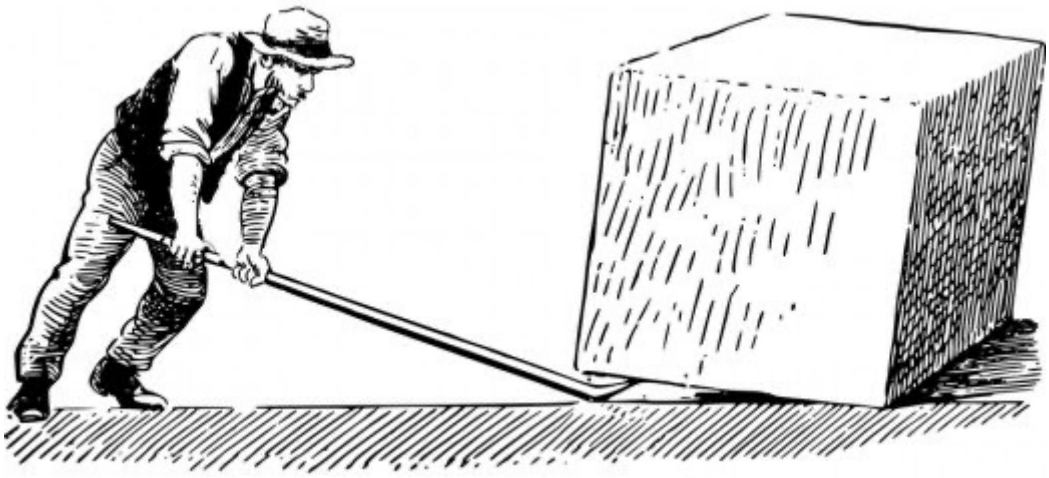
Complete the table below **4**

	Longitudinal waves	*Transverse waves*
Description	*Oscillations parallel to the direction of propagation.*	*Oscillations perpendicular to the direction of propagation.*
Example	*Sound*	*Water waves*

Marking Criteria	*Marks*
ONE mark for each of the following: • TWO correct definitions. • TWO correct examples.	4

Question 8 (3 marks)

Identify the simple machine present in the diagram below and explain how it works. **3**



First-Class Lever

Effort force is reduced compared to lifting the stack of hay without the simple machine.

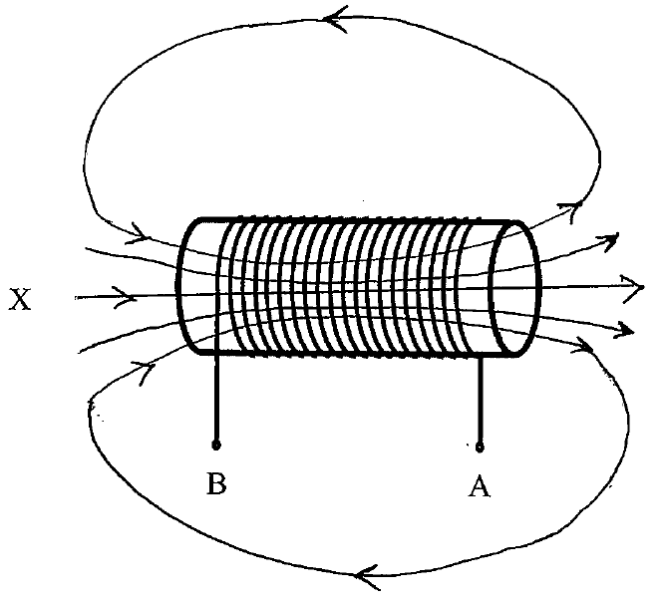
The amount of work done is the same, hence the multiplication of the distance from the fulcrum to the point where the force is applied with the magnitude of force should be equal for both the effort and load.

As the distance from the application of the effort force to the fulcrum is greater than the distance from the fulcrum to the application of the load force, the effort force is much less than the force which is applied to the load.

Marking Criteria	*Marks*
ONE mark for each of the following: • Identified that the main simple machine is a lever. • Identified the type of simple machine (First-Class Lever). • Explained how this simple machine has a mechanical advantage.	3

Question 9 (3 marks)

A group of students constructed a solenoid by winding electrical wire around a hollow cardboard tube as shown in the diagram below. **4**



An electrical current was passed through this solenoid from point A to point B.

(a) Draw the magnetic field lines around the solenoid in the diagram above. **2**

Marking Criteria	*Marks*
• Correctly illustrates magnetic fields lines around the solenoid with the correct direction.	2
• Correctly illustrates magnetic field lines around the solenoid with the incorrect or conflicting directions.	1

(b) Indicate the polarity of the magnetic pole generated at point X. **1**

South pole

Marking Criteria	*Marks*
• Correctly identifies the magnetic pole generated at point X from the direction of the magnetic fields drawn.	2

End of Section III

Section IV – Earth and Space

Name: .....

Class: .....

Mark: .....

Answer Grid for Multiple Choice Questions

1.	A O	B O	C O	D O
2.	A O	B O	C O	D O
3.	A O	B O	C O	D O
4.	A O	B O	C O	D O
5.	A O	B O	C O	D O

Part A – Multiple Choice Questions **5 marks**

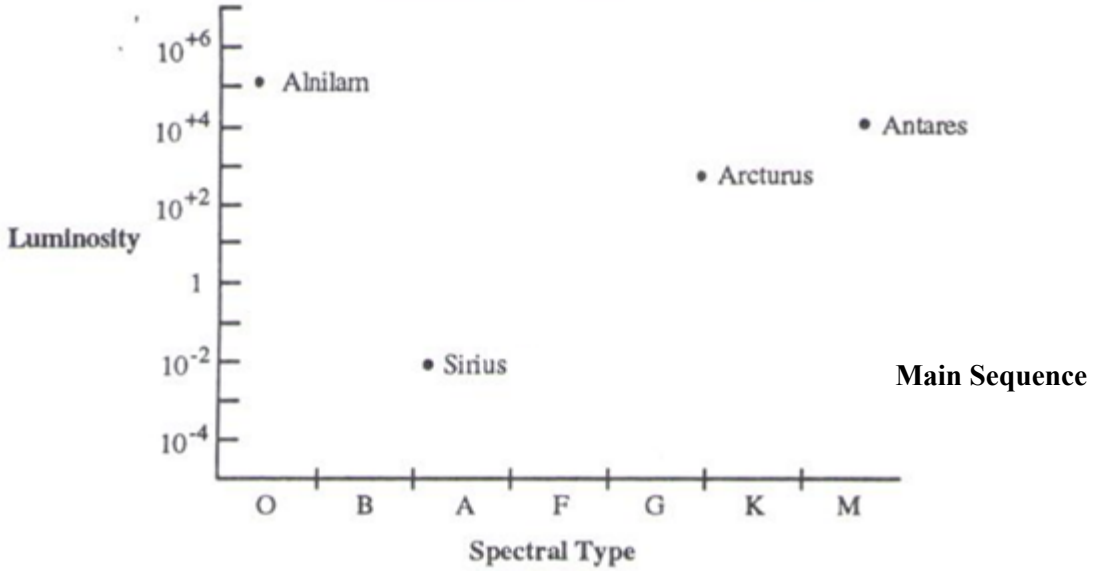
1. The crust and upper mantle are components of which of the following?

A Lithosphere
B Atmosphere
C Hydrosphere
D Asthenosphere

2. According to Wegener's model for plate tectonics, what evidence did glaciers leave for the existence of Pangaea?

A Striation patterns
B Sea-floor spreading.
C Glossopteris fossils
D Mid-ocean ridges

Use the information provided in the Hertzsprung Russell diagram to answer questions 3 and 4.



3. Which star in the diagram above has the greatest surface temperature?

A Sirius
B Antares
C Alnilam
D Arcturus

4. Which star in the diagram above is most similar to the Sun?

A Sirius
B Antares
C Alnilam
D Arcturus

5. The planet Mars is located approximately 228 million km from the sun.

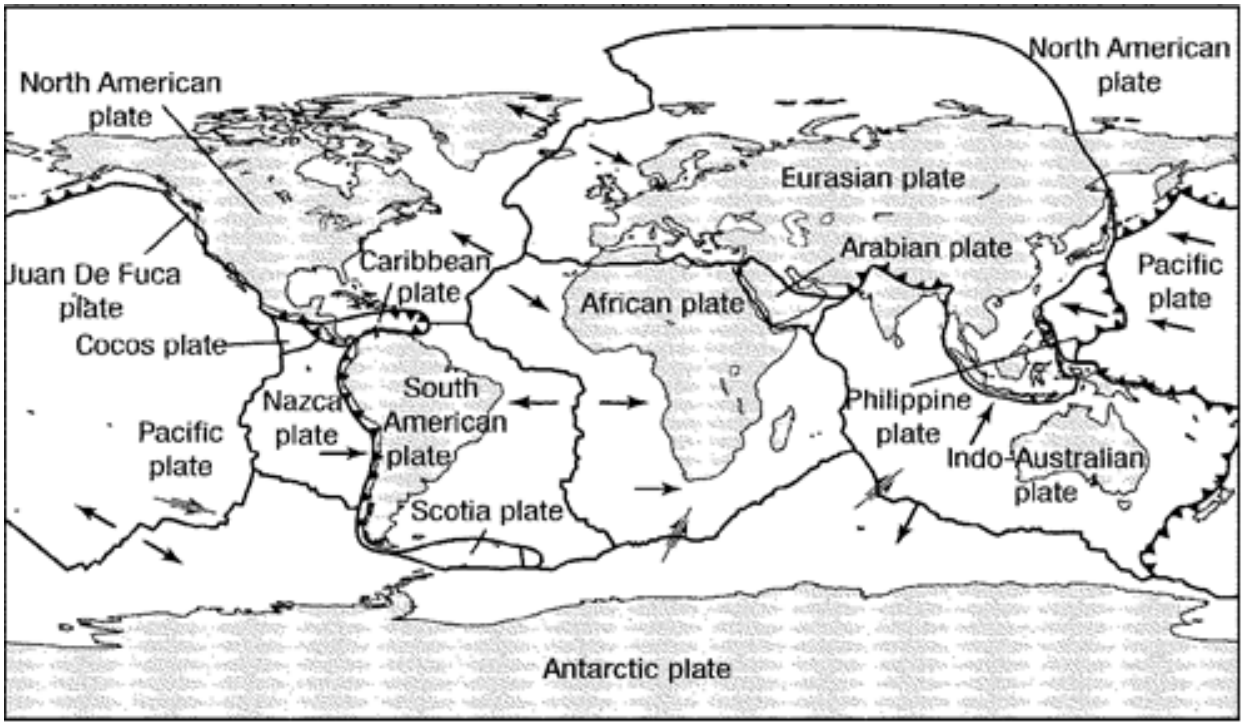
If the Sun were to release an intense burst of light in the form of a solar flare, how long would it take for this event to be observed on Mars given the speed of light is 3.0×10^8 ms^{-1}.

A 1.3 seconds
B 7.6 seconds
C 130 seconds
D 760 seconds

Part B – Extended Response Questions 15 marks

Question 6 (3 marks)

The diagram below illustrates the plate boundaries for the different continental plates.



(a) Identify the type of plate boundary between the South American and Nazca plates. **1**

Convergent

(b) Describe the features that would be observed at this plate boundary identified in (a) **2**

Subduction of the two plates would result in mountains, earthquakes, ocean trench, tsunamis, volcanoes.

Question 7 (2 marks)

Calculate how much energy is released in a nuclear reaction where 4.37 x 10^{-28} g of mass is lost using Einstein's mass-energy equivalence formula (**$E = mc^2$**). **2**

Show all necessary working.

$E = m X c^2 = 4.37 X 10^{-31} X (3 X 10^8)^2 = \underline{3.933 X 10^{-14} Joules}$

Criteria	Marks
Use correct values in equation with correct conversions to kg. Correct calculation with units	2
ONE wrong substitution in mass but method otherwise correct.	1

Question 8 (4 marks)

The stellar evolution of a star is dependent upon its mass during the main sequence phase of its lifespan. Predict the stellar evolution pathway for the following:

(a) A star with a similar mass to the sun. **2**

Stages Protostar ➔ Main sequence ➔ red giant ➔ nova ➔ white dwarf ➔black dwarf

Criteria	Marks
All correct in sequence well explained.	2
Insufficient details or terms to give full response.	1

(b) A star weighing up to 20 solar masses. **2**

Stages Protostar ➔ Main sequence ➔ red supergiant ➔ supernova ➔neutron star or black hole

Criteria	Marks
All correct in sequence well explained.	2
Insufficient details or terms to give full response.	1

Question 9 (2 marks)

Compare the density of rock samples obtained from the oceanic and continental crust. **2**

The average density of oceanic crust is 3.0 g/cm^3, while continental crust has an average of 2.7 g/cm^3

Criteria	Marks
All correct in reference to oceanic being higher in density. Accepted answer without detail.	2
Insufficient details or terms to give full response.	1

Question 10 (4 marks)

The Big Bang theory is the leading explanation for the origins of our universe.

Explain how the following pieces of evidence support the expanding universe model.

(a) Cosmological Red Shift **2**

Red Shift occurs when the red colours of the spectrum were shifted further to the red end of the light spectrum. Explanation – the reason being that those stars and galaxies are further away AND moving FASTER than those closer to us.

Hence this shift is evidence that the universe is still expanding at different velocities.

Criteria	Marks
Correct explanation of stars moving away at different speeds to account for variations in the shift. This implies evidence that the universe is still expanding but further out much faster.	2
Insufficient details or terms to give full good response understanding in answer as above	1

(b) Cosmic Microwave Background Radiation. **2**

CMBR is the residue energy left behind after the big bang event and 13 billion years of expansion. The residue energy now is microwave energy As evidence this implies that all that early high energy Gamma radiation has spread out as the universe expanded and now exists as low energy microwaves.

Criteria	Marks
Correct explanation of residue of the small Universe changing to a big expansion of the universe and the early GAMMA energy changing into todays low backgound microwaves energy. This implies evidence that the universe is still expanding and the residue energy remains today filling in all the universe.	2
Insufficient details or terms to give full good response understanding in answer as above	1

End of Examination